

CAPSTONE PROJECT DOCUMENTATION

SAP End-to-End Blueprint Data Analytics & Data Cloud

ABC Retail Pvt Ltd | Bengaluru, India

Field	Details
Project Type	SAP Blueprint — Data Analytics & Data Cloud
SAP Products	SAP Datasphere SAP Analytics Cloud (SAC) SAP BW/4HANA
Modules / Tools	Data Integration Data Modelling Live Dashboards Predictive
Domain	Electronics Retail — India
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Course / Batch	C_BCBDC
Submission Date	April 21, 2026

1. Problem Statement

ABC Retail Pvt Ltd is a fast-growing electronics retailer operating physical stores and an online portal across India. Prior to the SAP Data Analytics implementation, the company relied on fragmented data sources — spreadsheets, standalone POS systems, and disconnected ERP modules — resulting in the following critical business challenges:

- No single source of truth for sales, inventory, and financial data across Retail and Online channels.
- Business reports generated manually from multiple Excel files, causing delays of 3-5 days for weekly MIS reports.
- Inventory forecasting done manually, leading to frequent stockouts of high-demand SKUs and overstocking of slow-moving items.
- No real-time visibility into vendor performance, payment ageing, or procurement spend.
- GST reconciliation and e-invoice data maintained in separate tools, creating compliance risks.
- Management dashboards built in static PowerPoint slides with no live KPI tracking.
- Customer purchase behaviour data not analysed, resulting in missed cross-sell and upsell opportunities.
- IT team unable to integrate data from SAP ERP, POS terminals, and the online portal into a unified analytics layer.

SAP Datasphere (formerly SAP Data Warehouse Cloud) and SAP Analytics Cloud (SAC) were selected to build a unified, real-time data analytics platform for ABC Retail — connecting all operational data sources and delivering live dashboards, predictive insights, and self-service reporting.

2. Company Overview

Field	Details
Company	ABC Retail Pvt Ltd — Bengaluru, Karnataka, India
Industry	Electronics Retail (B2C) — Smartphones, Laptops, TVs, Appliances
Company Code	ABC01 Currency: INR Fiscal Year: April to March (Variant V3)
Plant	ABC_PLANT_01 — Main Warehouse, Bengaluru
Storage Location	ABCSL01 — Electronics Inventory Section
Sales Org	ABCSO01 — ABC India Sales
Sales Channels	RT (Retail Stores) + ON (Online Portal) Division: EL (Electronics)
Data Sources	SAP S/4HANA ERP POS Terminals Online Portal DB GST Portal
Analytics Platform	SAP Datasphere (Data Layer) + SAP Analytics Cloud (Presentation Layer)

3. Solution & Features

The SAP Data Analytics blueprint delivers a five-layer end-to-end data platform for ABC Retail. Data is ingested from all operational systems into SAP Datasphere, modelled into semantic business objects, and surfaced via SAP Analytics Cloud dashboards with predictive analytics and planning capabilities.

3.1 SAP Data Platform Architecture

Layer	SAP Component	Purpose for ABC Retail
1. Source Systems	SAP S/4HANA, POS, Online Portal	Operational transactions — sales, procurement, inventory
2. Data Integration	SAP Datasphere — Data Flows	ETL pipelines; real-time replication from S/4HANA via SLT
3. Data Modelling	Datasphere — Graphical Views	Semantic layer: sales model, inventory model, finance model
4. Analytics & BI	SAP Analytics Cloud (SAC)	Live dashboards, KPI tiles, story reports for management
5. Predictive & Planning	SAC Predictive Scenarios	Demand forecasting, sales planning, inventory optimisation

3.2 Key Data Models in SAP Datasphere

Model Name	Type	Key Measures / Dimensions
Sales Analytics Model	Analytical Dataset	Revenue, Units Sold, Channel (RT/ON), Product Category, Date
Inventory Model	Graphical View	Stock Qty, Reorder Level, Turnover Rate, Plant, Storage Loc
Procurement Spend Model	Analytical Dataset	PO Value, Vendor, Payment Status, Ageing Buckets
Customer Behaviour Model	Analytical Dataset	Purchase Frequency, Basket Size, Channel Preference, Region
GST Compliance Model	Graphical View	CGST, SGST, IGST, Invoice No, GSTIN, Filing Period
Financial Overview Model	Analytical Dataset	Revenue, COGS, Gross Margin, AR/AP Balances, Cash Flow

3.3 SAP Analytics Cloud — Dashboard Suite

Dashboard	Audience	Key KPIs Displayed
Executive Overview	CEO / CFO	Total Revenue, Gross Margin %, Cash Position, Top 5 SKUs
Sales Performance	Sales Manager	Daily Sales by Channel, Target vs Actual, Region Heat-map

Dashboard	Audience	Key KPIs Displayed
Inventory Intelligence	Operations Head	Stock Levels, Stockout Alerts, Slow-moving Items, Days on Hand
Procurement Analytics	Purchase Manager	Spend by Vendor, PO Cycle Time, GR/IR Pending, Savings
GST Compliance	Finance / Tax Team	Output GST, Input Tax Credit, GSTR-1 vs GSTR-3B Variance
Demand Forecast	Planning Team	Next 30/60/90-day forecast per SKU using SAC Smart Predict

4. SAP Configuration Steps

All configuration spans SAP Datasphere (tenant setup, spaces, data integration) and SAP Analytics Cloud (connections, models, stories). Steps must be executed in the sequence below.

#	Step	Tool / Path	Details
1	Create Datasphere Tenant & Space	Datasphere Admin Console	Create Space 'ABC_RETAIL_SPACE'; assign storage quota; set INR currency.
2	Configure S/4HANA Live Connection	Datasphere > Connections > New > SAP S/4HANA	OData / SLT replication connection to ABC01. Enable delta replication.
3	Replicate Source Tables	Datasphere > Data Builder > Replication Flow	Replicate MM, SD, FI tables: VBAK, VBAP, MKPF, MSEG, BSEG, LFA1, KNA1.
4	Build Graphical Views	Datasphere > Data Builder > Graphical View	Join replicated tables to create Sales, Inventory, Finance Models.
5	Create Analytical Datasets	Datasphere > Data Builder > Analytical Dataset	Expose models with measures, dimensions, and hierarchies.
6	Define Business Entities	Datasphere > Business Builder	Create entities: Products, Customers, Vendors, Fiscal Calendar V3.
7	Connect SAC to Datasphere	SAC > System > Connections > SAP Datasphere	Live connection using tenant URL; enable SSO with SAP BTP Identity Service.
8	Build SAC Stories / Dashboards	SAC > Stories > New > Canvas	Create 6 dashboards (Section 3.3) using Charts, KPI Tiles, Geo Maps.
9	Enable Smart Predict — Demand Forecast	SAC > Predictive Scenarios > Time Series	Train model on 24 months of sales data; forecast next 90 days per SKU.
10	Schedule Refresh & Alerts	SAC > Calendar & Datasphere > Task Chains	Hourly delta replication; daily full refresh at 2 AM; email alerts on KPI breach.

5. Unique Points of This Project

Unique Point	Description
Omni-Channel Data Unification	Sales from Retail (RT) and Online (ON) channels merged into a single semantic model — enabling channel-wise P&L for the first time.
India-Specific Compliance Layer	Dedicated GST Compliance Model covering CGST, SGST, IGST with GSTR-1 vs GSTR-3B variance detection built into the dashboard.
SAP Datasphere Semantic Layer	Business Builder entities (Products, Customers, Vendors, Fiscal Calendar V3 Apr-Mar) make reports self-service for non-technical users.
AI-Driven Demand Forecasting	SAC Smart Predict trained on 24 months of sales history to generate 90-day SKU-level demand forecasts — reducing stockouts.
End-to-End Lineage & Governance	Datasphere Data Catalogue tracks data lineage from source S/4HANA tables to final SAC dashboard tiles — full audit trail.
Real-Time Live Connection	SAC connects to Datasphere via Live Connection — dashboards reflect S/4HANA transactions within minutes, no overnight batch.

6. Future Improvements

Improvement Area	Description
SAP HANA Cloud Integration	Migrate data warehouse to SAP HANA Cloud for in-memory query performance on large datasets.
Joule AI Copilot in SAC	Enable SAP Joule natural-language interface — managers query dashboards in plain English (e.g., 'Show me top 10 SKUs this week').
SAP CX / CRM Integration	Integrate customer purchase history from SAP Commerce Cloud for 360-degree customer analytics and loyalty program insights.
Embedded Analytics in S/4HANA	Deploy SAC embedded analytics tiles directly inside S/4HANA Fiori — procurement and finance users see KPIs in their workflow.
Sustainability Reporting	Add carbon footprint tracking per product category for ESG compliance using SAP Sustainability Footprint Management.

Conclusion

This project presents a complete SAP Data Analytics and Data Cloud blueprint for ABC Retail Pvt Ltd. The solution leverages SAP Datasphere for unified data integration and semantic modelling, and SAP Analytics Cloud for real-time dashboards and AI-driven forecasting. The architecture addresses all identified pain points — from fragmented reporting and manual MIS to GST compliance risk and inventory blind spots — transforming ABC Retail into a data-driven, insight-ready organisation fully aligned with SAP's Intelligent Enterprise vision for Indian retail businesses.

References: [SAP Help Portal \(help.sap.com\)](https://help.sap.com) | [SAP Datasphere Documentation](#) | [SAP Analytics Cloud Learning Hub](#) | [SAP India \(sap.com/india\)](https://sap.com/india) | [SAP BW/4HANA Guide](#) — SAP Press